

Research

TXtreme: transformer-based extreme value prediction framework for time series forecasting

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Abstract

Time Series Forecasting (TSF) is crucial in various real-world applications such as climate forecasting and electricity demand prediction. Unlike traditional datasets, time series data points are influenced by their past values, necessitating specialized techniques to model these sequential dependencies specifically addressing non-linear patterns, abrupt changes, and outliers. The latest advancements have significantly enhanced TSF using machine learning and other methods. However, forecasting extreme events remains challenging. Extreme values, although rare, have significant real-world impacts such as heavy rainfall, fluctuations in electricity demand, and traffic surges. This paper proposes a TXtreme framework that uses Long-Short memory network, feed-forward neural network, and transformer to improve time series forecasting under extreme values. The model also uses statistical methods to explain the distribution of time series values. Extensive experiments are conducted using datasets from different domains to show the robustness of the proposed methodology. Results, derived by testing TXtreme on five datasets of different domain, indicate that TXtreme significantly outperforms state-of-the-art methods in time series forecasting, with improvements of 5–25% in root means squared error or mean absolute error. The proposed framework enhances TSF capabilities and ensures better generalization ability in extreme event forecasting, potentially leading to improved decision-making in critical applications.

Article Highlights

- Approach to focus on extreme values in time series forecasting.
- TXtreme is a hybrid approach for improving forecasting accuracy.
- The Gaussian Mixture model is used to create an extreme value indicator.

Keywords Extreme value theory · Transformer · Mixture models · Hybrid model · Time series forecasting

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1 Introduction

Time series forecasting (TSF) has been applied extensively in various real-world scenarios, including precipitation prediction [1], wind speed estimation [2, 3], stock market forecasting [4], sales predictions [5], traffic flow estimation [6], weather forecasting [7], and electricity demand prediction [8], among others. The sequential nature, autocorrelation, and inherent patterns of time series data require specialized forecasting techniques [9]. Consequently, over the past decades, researchers have focused on developing advanced models and methods capable of capturing the non-linear relationships present in time series data [10]. Recently, data driven approaches have gained considerable traction in addressing data-driven problems, including TSF [11]. Deep neural networks (DNNs) are becoming a superior alternative to the traditional statistical methods for time series forecasting, as they can model multiple time series concurrently. The flexibility of DNNs, which allows them to learn from data without requiring prior assumptions, offers substantial advantages when working with real-time time series data [12]. With the growing research interest in deep learning, neural network architectures specifically designed for sequential data, such as recurrent neural networks (RNNs) [9] and their variants, as well as Transformer models [13], have demonstrated significant improvements in TSF by capturing long-term temporal dependencies and learning complex non-linear functions. These models excel in both point forecasting and multi-horizon prediction, as they effectively capture intricate patterns in the data, enabling better adaptation to sudden changes and outliers. Furthermore, ensemble and hybrid models, which integrate multiple forecasting methods, have been proposed to further improve the accuracy of predicting extreme events [14].

While progress has been made, accurately predicting rare and extreme occurrences in time series data continues to be a complex and evolving field of study [4]. This underscores the importance of developing tailored methods specifically designed to forecast time series that include unusual or extreme data points. Extreme values in time series, representing rare but impactful events like surge in electricity demand or natural disasters, pose unique challenges to forecasting accuracy. These extreme values can significantly skew predictions if not properly accounted for, highlighting the need for robust models and techniques that specifically address the complexities of handling extreme values in forecasting. In simple terms, extreme values are those that are much larger or smaller than the typical range of the time series. The data distribution of such extreme values remains different compared to those of normally distributed data in time series like heavy rainfall and floods, abrupt changes in stock indices, or substantial traffic surges. In this paper, we present a customizable framework for time series forecasting that emphasizes modeling both extreme and normal values to minimize overall prediction error. The main objectives of this research are listed below,

- Develop a method that effectively reduces the impact of extreme values on prediction error, even when the occurrence of such extreme values is relatively sparse in comparison to normal values within the time series.
- In a unique effort normal time series values are being estimated using the sequence learning model LSTM and time series with extreme events are being modeled by the transformer.
- Through an extensive series of experiments on real-world datasets, we demonstrate that the proposed model delivers high accuracy in predicting extreme events and performs comparably to leading state-of-the-art time series forecasting models.
- It is also shown that incorporating an extreme event classification model can significantly reduce prediction errors in extreme event forecasting tasks.

The primary objectives of this research are to develop a robust prediction method that effectively mitigates the impact of extreme values on prediction errors, even when such values are scarce. This involves employing a dual modeling strategy where normal time series values are estimated using Long Short-Term Memory (LSTM) networks, while extreme events are modeled using transformers. Extensive experiments across diverse real-world datasets will demonstrate the TXtreme framework's high accuracy in predicting extreme events, positioning it competitively among leading time series forecasting models. Additionally, a specialized extreme event classification model using a Feed-Forward Neural Network will significantly reduce prediction errors related to extreme events, enhancing overall forecasting accuracy.

The paper has the following sections. Section 2 covers the related work, Sect. 3 discusses the preliminary concepts necessary for the methodology, Sect. 4 details the proposed methodology, Sect. 5 outlines the experimental setup, Sect. 6 presents the experimental results, and finally, Sect. 7 provides the conclusion.

2 Related work

Time series forecasting plays a vital role in numerous domains, including renewable energy, air quality management, demand forecasting, and weather prediction. Traditional statistical methods like ARIMA (Auto-regressive Integrated Moving Average) have remained widely used for their accuracy, efficiency, and ease of use [9]. However, recent studies have shown that machine learning methods, including recurrent neural networks (RNN) and deep learning approaches like LSTM, are gaining traction in time series forecasting [15, 16]. These methods have demonstrated strong performance, particularly in situations involving larger sample sizes [15]. In the field of renewable energy forecasting, deep learning models such as LSTM have been employed for short-term wind speed prediction, showing promising results when compared to traditional statistical models [17, 18]. Similarly, for solar irradiation forecasting, a novel approach combining Gramian Angular Field and Convolutional LSTM networks has shown improved forecasting performance despite using a small dataset [19]. A hybrid approach that integrates machine learning techniques with statistical methods has also been suggested for forecasting solar power generation [20] and dissolved oxygen level prediction [21], demonstrating improved accuracy over standalone models. A combined interval forecasting method proposed in [22], which integrates three artificial intelligence models (LSTM, BiLSTM, and GRU) with a statistical model (ARIMA), leveraging interval prediction mechanisms for a reliable forecasts amidst uncertainty in wind energy generation. In the field of air quality prediction, a Markov-weighted fuzzy time-series model utilizing an optimal partitioning technique has been created for forecasting air pollution, surpassing the performance of traditional fuzzy time-series models and conventional approaches [23]. Furthermore, the integration of wavelet transform with convolutional neural networks has proven effective in identifying patterns within rainfall time series, leading to satisfactory forecasting outcomes [24]. In the healthcare domain, a comparison of statistical, neural, and ensemble methods for predicting patients' weekly average spending on pain medications has demonstrated that ensemble models outperform individual approaches. [25]. Overall, the literature indicates the field of time series forecasting is shifting towards the adoption of machine learning or deep learning methods, showcasing competitive performance in various domains. While statistical models remain popular for their simplicity, the advancements in neural network architectures offer promising alternatives for efficient forecasting.

Handling extreme values in time series is often overlooked by deep learning and machine learning models. The paper [26] highlights the challenges of diverse numerical routines, aiming to compare and discuss practical issues often overlooked in theoretical discussions. This oversight primarily arises from the significant imbalance in the data, where extreme events are rare compared to more common occurrences. Deep models, designed to minimize overall error, tend to prioritize frequent patterns, leading to underfitting (overlooking extreme events) when dealing with extreme values. Extreme events require specialized models that can manage these imbalances effectively. Various approaches in the literature to address the problems of time series are summarised in Table 1. Table 1 provides a summary, key contribution, and limitations of the approach which indicates requirements for developing a forecasting method that specifically addresses extreme values and also be effective in overall forecasting.

3 Preliminaries

3.1 Defining TSF

Consider (N) as the total number of independent sequences. Each sequence is a pair of X and corresponding y as given in Eq. 1. If T is the input length and H is the forecasting horizon, X can be given by $X = \{x_1, x_2, x_3, \dots, x_T\}$ and y can be represented as $y = \{y_1, y_2, y_3, \dots, y_H\}$. For the i^{th} sequence, the time series data for the machine learning model can be described as:

$$\{(X^{(i)}, y^{(i)}) | i = 1, 2, 3, \dots, N\} \quad (1)$$

The objective of time series forecasting model development is to establish the mapping between X and y such that it minimizes the absolute difference between prediction O^i and actual value y^i at any time t given by Eq. 2.

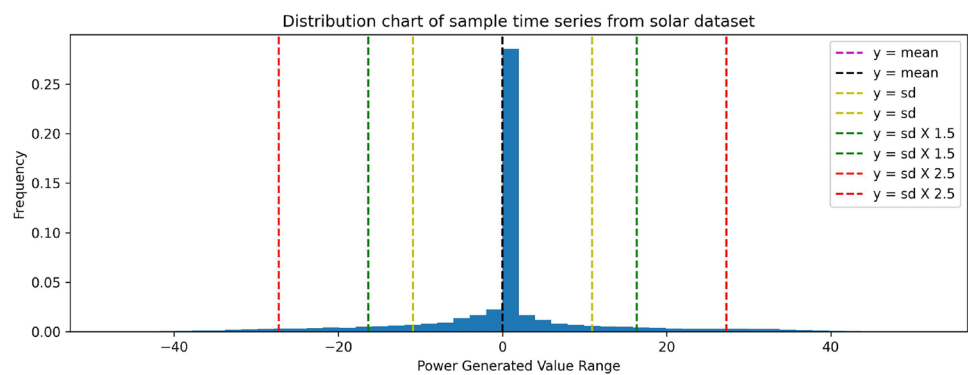
$$\min \sum_{(i=1)}^H |O_t^i - y_t^i| \quad (2)$$

In this research time series from various domains are considered for developing a forecasting framework. The dataset from each domain contains multiple time series and a major contribution in the performance of time series forecasting is by forecasting normal values. However, there are extreme values that are a minority compared to normal values and

Table 1 Comparison of time series forecasting approaches

Paper	Approach	Key contributions	Limitations
[27]	Extreme Value Theory with similarity measure	Predicts extreme events, but computationally inefficient and limited to seen patterns	Computational cost, limited generalizability
[28]	Neural network	Predicts complicated trends during high-variance segments	Lack of uncertainty information, limited generalizability
[28]	Bayesian Neural Network with Monte Carlo dropout	End-to-end time series forecasting with uncertainty estimation	No specific focus on extreme values
[29]	Memory-efficient framework	Targets extreme events, integrates exogenous series and uses quantile loss	Requires expert analysis for stored extreme events
[30]	Generalized Extreme Value Loss (GEVL)	Uses heavy-tailed distribution kernels for loss estimation	No specific focus on extreme events
[31]	Causal tail coefficient	Detects causal relationships between extreme events	Limited to heavy-tailed time series
[32]	Variational disentangled external classifier (VIE)	Classifies non-stationary time series data with extreme events	Only addresses right extremes
[33]	Random threshold estimators	Shows similar limit distributions for random and deterministic thresholds	Slightly worse finite-sample performance for deterministic thresholds

Fig. 1 Distribution after differencing for sample time series from Solar dataset



also have a significant portion in the performance of forecasting. So in this scenario, extreme values must be precisely forecasted along with normal values to reduce overall error and avoid disastrous events to some extent.

3.2 Extreme values

The imbalance in time series data is due to extreme values, which generally follow a different distribution than normal values in time series. Equations 3, 4 and 5 represents it formally, where β_1 and β_2 are upper threshold and lower threshold respectively. Consider some time series $X_{(1:T)} = [x_1, x_2, \dots, x_T]$. Now for any value in time series $X_{(1:T)}$, at time t , consider an indicator sequence $I_{(1:T)} = [i_1, i_2, \dots, i_T]$:

$$i_t = 1, \text{ if } (x_t > \beta_1) \quad (3)$$

$$i_t = 0, \text{ if } (x_t \in -\beta_1, \beta_2) \quad (4)$$

$$i_t = -1, \text{ if } (x_t < -\beta_2) \quad (5)$$

Values in indicator sequence $I_{(1:T)}$ give a difference between extreme values and normal values.

3.3 Extreme values vs outlier

While the terms “outliers” and “extreme values” are often used interchangeably in time series analysis, they can have subtle differences in meaning and interpretation. Table 2 lists a pointwise comparison of extreme values with outliers. Outliers in time series are data points that are significantly high or low from the overall range. They can be identified using statistical measures like Dixon’s test [34], Grubb’s Test [35] and z-scores [36]. Outliers are values recorded due to measurement errors, data entry mistakes, or changes in underlying processes. Outliers are significantly low in the datasets and are not points of interest so they can be treated by removal. On the other hand extreme values though very little are in considerable proportion due to which they can not be removed. Figure 1 shows a sample time series distribution chart from the Solar dataset in which values beyond red lines are outliers while values between red and green lines are extreme.

Table 2 Comparing outliers and extreme values

Feature	Outliers	Extreme Values
Definition	Data points with very large or small values from the overall pattern and recorded due to some error in data collection process.	Data points in the tails of the distribution, often representing the highest or lowest values.
Causes	Measurement errors, data entry mistakes, unusual events, and changes in processes	Rare events with potentially high impact
Handling	Removal, correction, winsorization, robust methods	Modeling using extreme value theory to estimate probabilities and magnitudes.

Various techniques proposed in the literature to model extreme values such as Generalized Pareto Distribution [37] and generalized Extreme value theory (EVT) [38] provide a framework for modeling these values, often using distributions. Extreme events can have significant consequences, such as financial losses in finance or widespread damage to environmental science. Extreme values are recorded due to extreme events and extreme events can be classified based on the type of uncertainty and error of forecasting as given in [39].

3.4 Limitations of deep learning models (DNNs)

Deep neural networks (DNNs) can struggle with extreme events for several reasons. To mathematically justify why deep neural networks (DNNs) can suffer from the extreme event problem, consider the following explanation.

3.4.1 Data imbalance and rare events

Extreme values are often underrepresented in training data, leading to a model that is not well-equipped to handle them. When extreme values are underrepresented, the model has fewer examples to learn from, making it difficult to accurately capture its magnitude and pattern. This is especially problematic in time series where most data follows a normal pattern, and extreme values are sparse. Bayesian methodology can be one of the most simplest to express the probability of an extreme values in time series [40]. The probability of extreme values can be given by Eq. 6.

$$P(E|X_{1:T}) = \frac{P(X_{1:T}|E)P(E)}{P(X_{1:T})} \quad (6)$$

Where, E indicates an extreme value in the time series. $X_{1:T}$ represent the observed time series data up to time T . The posterior probability of an extreme value occurring is given by $P(E|X_{1:T})$ for the observed data. $P(E)$ is the prior probability of an extreme event occurring, based on prior knowledge or historical data. $P(X_{1:T}|E)$ is the likelihood of observing the time series data $X_{1:T}$ given that an extreme event occurs. $P(X_{1:T})$ is the evidence, which represents the total probability of observing the data $X_{1:T}$, regardless of whether an extreme value occurs. For example, to determine the probability of an extreme rainfall event E based on observed rainfall data over the past T time steps, $X_{1:T}$. Using Eq. 6, it can be given as below.

- Prior Probability $P(E)$: The probability of an extreme rainfall day, estimated from historical data (e.g., 5% of days in the past have experienced extreme rainfall).
- Likelihood $P(X_{1:T}|E)$: The probability of observing the rainfall data $X_{1:T}$ under the assumption that an extreme event has occurred. This distribution could be characterized by using models tailored to extreme events, though GEV is specifically applicable to extremes rather than the complete time series. $P(X_{1:T}|E)$ is not being modeled directly using GEV. Instead, it is decomposed into components that relate to extreme behavior and Components that relate to normal behavior.
- Evidence $P(X_{1:T})$: The total probability of observing the data $X_{1:T}$, which serves as a normalizing factor, ensuring that posterior probabilities sum to 1.

In this context, the posterior probability $P(E|X_{1:T})$ indicates the updated probability of an extreme event, conditioned on the observed rainfall data up to time T .

3.4.2 Model complexity and over-fitting

DNNs are complex and may overfit normal patterns while missing on extreme values. DNNs can easily memorize frequent patterns, but they might not learn the nuanced characteristics of rare events due to their complexity and the imbalance in data representation. This results in an output that performs well on common patterns and on extreme values performance is below average. Consider a DNN with parameters θ . The loss function $L(\theta)$ minimized during training can be expressed as in equation 7:

$$L(\theta) = \frac{1}{N} \sum_{i=1}^N \ell(y_i, f(X_i; \theta)) \quad (7)$$

Where, ℓ is a loss for a single instance. For rare events, $y = y_{rare}$ contributes minimally to the sum, causing the model to not learn these rare patterns effectively, leading to overfitting on frequent patterns and underfitting on rare events.

3.4.3 Sensitivity to outliers

DNNs might not generalize well to extreme events if those events deviate significantly from the training data's distribution. Extreme events can be seen as outliers. The presence of these outliers can skew the model's training process, as the loss function becomes heavily influenced by these rare points, leading to overfitting on these specific instances and poor generalization. As in Eq. 8, If y_i represents an extreme event, it can act as an outlier. Consider the gradient of the loss function $\ell(y_i, f(X_i; \theta))$:

$$\nabla_{\theta} \ell(y_i, f(X_i; \theta)) \quad (8)$$

For outliers, the gradient might be disproportionately large, causing the model to over-adjust for these rare points during training, which can destabilize learning and lead to poor generalization.

4 Methodology

TXtreme framework is divided into three parts. Details of each are discussed in this section.

4.1 Proposed model architecture

Figure 2 shows the first section about the process of preparing data for the model training. Figure 4 shows how the data is fed to each model and is all about modal training, Finally, Fig. 4 also depicts the process of preparing a forecast from the outputs of the middle section.

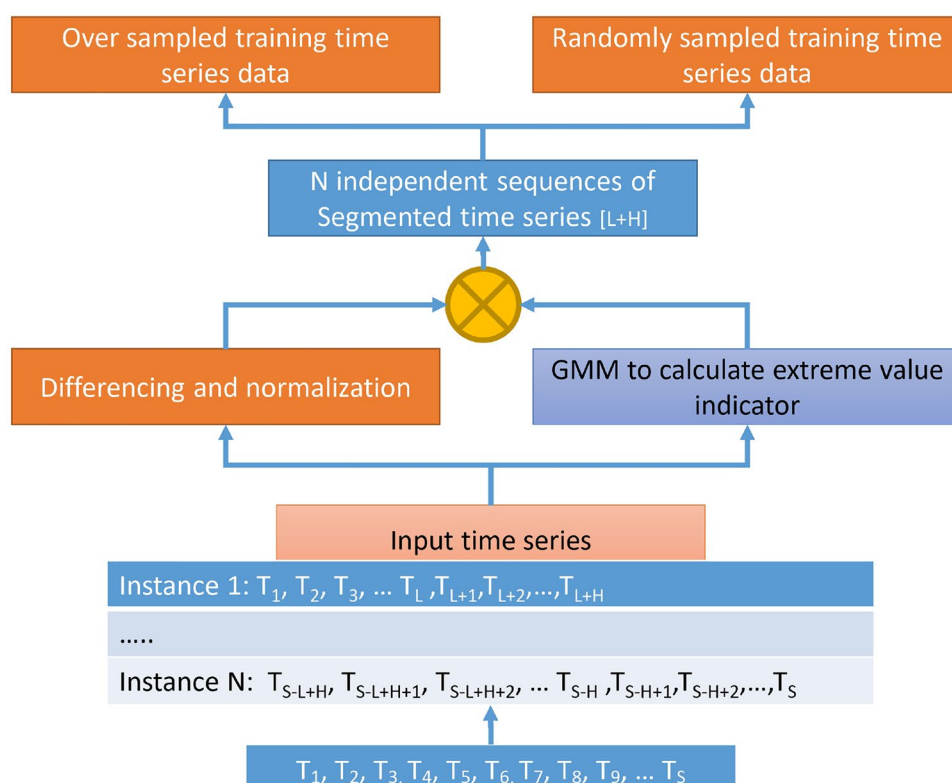
4.1.1 Preparing data

This is the initial step for model development. First step in preporcessing is normalization and detrending to convert the input data into a stationary time series. Next, we calculate the extreme value indicator using a Gaussian Mixture Model (GMM). GMM model uses a weighted average of 3 components for calculating extreme values indicator sequence for all time series sequences. This indicator is then concatenated with the input time series data to provide a comprehensive dataset. To create training samples, a sliding window approach is used where each window of L time steps is treated as an input sequence, followed by the target values for the next H steps. Specifically, the window slides one time step forward at each iteration, capturing overlapping sub sequences and allowing the model to learn from multiple segments of historical data. For example, the first window would cover time steps 1 to L, and the target would be steps L+1 to L+H; the next window would then shift to cover time steps 2 to L+1, with the target now covering steps L+2 to L+H+1, and so on. This method ensures the model learns temporal patterns effectively by leveraging all possible sequences in the time series. Each training sample thus consists of past data over L steps across all sensors and the target values for the next H steps. After preparing the data, it is sampled using two strategies. The first strategy is random sampling, which selects N samples from the total set of data to train a normal forecaster. The second strategy involves over-sampling the data to feed to the transformer model. For the classification model, data is prepared by augmenting it with extreme value indicators. These indicators, created with GMM and concatenated with L, enable the model to account for and predict extreme events more accurately, enhancing overall forecasting performance.

4.1.2 Model training

Three models namely Normal Forecaster, Extreme Forecaster, and extreme value classifier are trained as below.

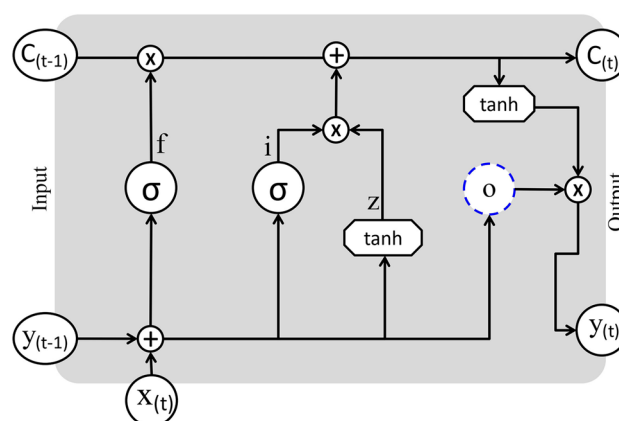
Fig. 2 Data Preprocessing of the framework (S indicates total length of the series, L indicates look-back period, and H indicates forecasting horizon)



Normal Forecaster: It is an LSTM-based model for forecasting normal values in time series that involves several key components to effectively capture temporal dependencies and produce accurate predictions. It begins with an input layer that is of 100 units of LSTM Cells followed by a batch normalization layer. The LSTM model contains a total 4 layers each of 100 cells of LSTM units and followed by batch normalization layers. The output from the LSTM layers is fed into a dense (fully connected) layer, which maps the learned features to the final output. This model uses a modified version of the LSTM cell called NOA-LSTM proposed by [41]. NOA-LSTM is depicted in the Fig. 3 The output layer provides the forecasted value for the specified forecasting horizon. The forecasting horizon is typically determined by the domain and business context. However, to evaluate the performance of the TXtreme model consistently across all datasets, we set a common forecasting horizon of 72 (3 days ahead). This choice allows us to focus on the characteristics of the time series data rather than domain-specific requirements, ensuring a fair comparison of the model's effectiveness in capturing both normal and extreme values.. The model is compiled using Huber loss as discussed in Sect. 4.3 and Adam(Adaptive Momentum) [42] optimizer.

Extreme Forecaster: The Extreme Value Forecaster leverages the Transformer architecture [43] for time series forecasting, specifically designed to focus on rare events or extreme values. Unlike traditional models, the Transformer's attention

Fig. 3 NOA-LSTM Cell architecture



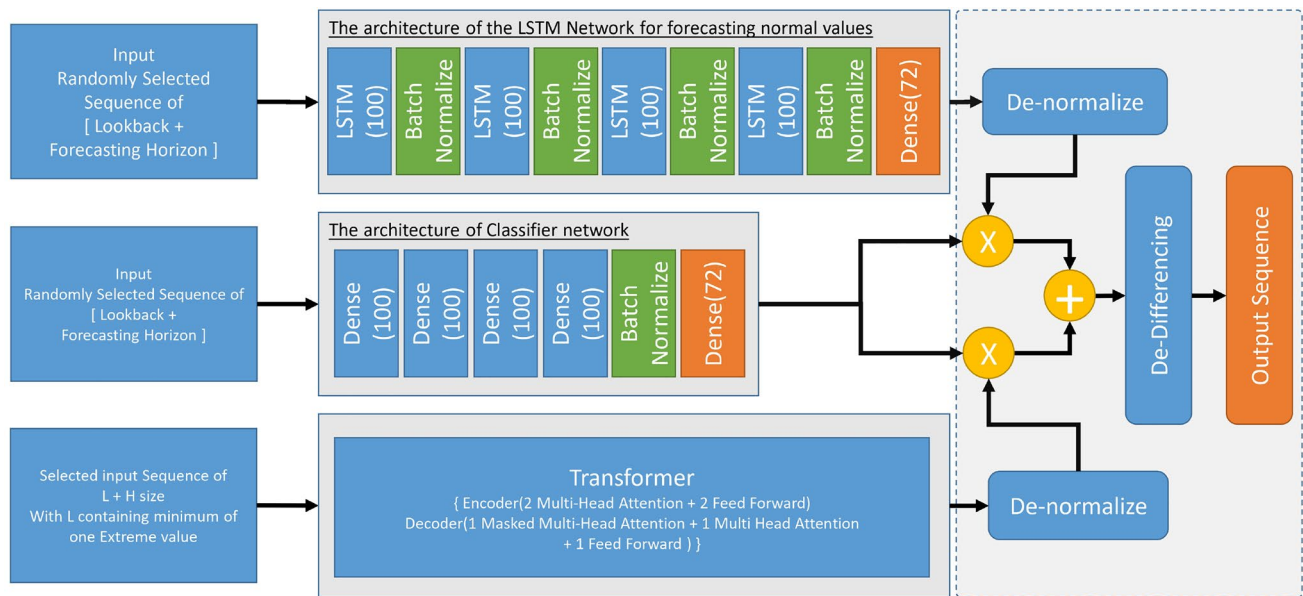


Fig. 4 Model Training process of the framework

mechanism allows it to dynamically adjust its focus based on the input sequence, making it particularly adept at capturing and predicting extreme values. This capability is crucial in scenarios where accurately forecasting rare events is paramount, such as in financial markets, anomaly detection, or natural disaster forecasting. The Extreme Value Forecaster's architecture consists of multiple attention layers that enable it to effectively learn and model the complex patterns associated with extreme values, leading to improved forecasting accuracy and robustness in extreme event prediction.

Extreme value classifier: The Extreme Value Classifier is constructed using a Feed Forward Deep Neural Network (DNN) to identify and classify extreme values in time series data. Unlike traditional classifiers, the DNN architecture allows for learning intricate patterns and relationships within the data, enabling it to effectively distinguish between normal and extreme values. The DNN consists of multiple hidden layers, each containing a large number of neurons, which enables it to learn complex representations of the input data. By training on a diverse range of time series data containing normal and extreme values, the Extreme Value Classifier can accurately identify and classify extreme events (Fig. 4).

4.1.3 Preparing forecast

Output of LSTM and Transformer model are denormalized and multiplied with output of classifiers in corresponding elements. These applied outputs are then combined with the summation of corresponding elements. This process results in normal outputs from the LSTM model as normal values in time series while extreme output by transformers from all extreme values. The TXtreme framework uses a selective integration approach, routing sequences to either an LSTM for normal values or a transformer for extreme values based on a classifier's output. The final forecast is generated by merging outputs after denormalization and de-differencing. This hybrid approach uses specialized models for different data types, optimizing accuracy compared to traditional ensemble methods.

4.2 Data segmentation and sampling for training

The TXtreme models utilize a sequence of L historical values to estimate the next H values, where $L + H$ equals the total length of the analyzed time series segment. Although the quantity of extreme values depends on the chosen threshold, it remains relatively small compared to normal values. In our experiments, we set $L = 360$, $H = 72$, and the total time series count N to vary depending on the dataset. Extreme values portion ranges between 0.08% to 4.08% percent of the total values across 5 datasets used in this research paper. Given this imbalance, an effective sampling strategy is essential for training. It was observed that while oversampling enhances the prediction of extreme events, it has a detrimental effect on overall accuracy. To mitigate this, a two-stage sampling strategy was implemented in the TXtreme model. In the first stage, random subsections of length $L + H$ were sampled from the time series data, ensuring they did not overlap with

the test and validation sets. The validation and test sets comprised 10 percent randomly selected sections of length L , while the remaining data was allocated for training. In the second stage, stratified sampling was used to balance regions with and without extreme values, allowing the Transformer and Classifier models to oversample a controlled percentage of samples that included at least one extreme event within the prediction window.

4.3 Loss function for training

To improve forecasting accuracy and mitigate sensitivity to extreme values in time series forecasting, specialized loss functions must be employed. Absolute loss functions create problems when being differentiated at zero. Also, quadratic loss functions are very sensitive to extreme values which again imposes distortions in model training. Huber loss is made up of the required characteristics of mean absolute error (MAE) and mean squared error (MSE) [30] which can be given as 9. Also, Huber loss is less affected by outliers than MSE, and unlike MAE, it is differentiable at zero, which is beneficial for optimization algorithms.

$$L_{\delta}(a) = \begin{cases} \frac{1}{2}a^2, & \text{if } |a| < \delta \\ \delta\left(|a| - \frac{1}{2}\delta\right), & \text{otherwise} \end{cases} \quad (9)$$

Here, a is the residual (difference between predicted and actual values), and δ is a threshold parameter to determine points where the loss function transitions from quadratic to linear.

5 Experimental setup

This section outlines the various datasets utilized in the experiments, including the preprocessing steps applied to them. It also provides details on the procedures for evaluation of the models, as well as the benchmarks employed for comparative analysis. The TXtreme framework is implemented using TensorFlow 2.8, and all experiments are conducted on a single NVIDIA Quadro 5000 16 GB GPU.

5.1 Dataset details

Our dataset contains datasets from various domains as listed in Table 3. The experiment dataset varies in domain, frequency, magnitude, distribution, and observation length. The first dataset, about solar energy, contains 137 series with a minimum length of 52,560 data points, collected at a frequency of every 10 min. The third dataset, related to San Francisco traffic, comprises 862 series with a minimum length of 17,544 data points, and collected hourly. The fourth dataset, concerning electricity, includes 321 series with a minimum length of 26,304 data points, and like the San Francisco traffic dataset and recorded hourly. The fifth dataset, on web traffic, is extensive with 145,063 series, and each series has a minimum length of 803 data points and is recorded daily. These datasets collectively offer a rich resource for research across energy, transportation, and web analytics catering to various analytical needs and frequencies of data collection.

5.2 Baseline methods

Baseline methods are state-of-the-art time series forecasting models. These models, include Feedforward Neural Networks (FFNN), DeepAR, N-BEATS, WaveNet, and Transformer.

Table 3 Dataset statistics

Dataset	Usage in research	Domain	No. of Series	Length	Frequency
Solar	2023 [44], 2022 [45], 2018 [46]	Energy	137	52,560	10 Min
San Francisco Traffic	2023 [47], 2022 [48], 2021 [48], 2020 [49], 2019 [50]	Transport	862	17,544	Hourly
Electricity	2021 [51], 2021 [9], 2021 [52], 2019 [53], 2018 [54]	Energy	321	26,304	Hourly
Web Traffic	2021 [51], 2021 [9], 2021 [52], 2019 [53], 2018 [54]	Web	145,063	803	Daily

Feedforward Neural Network (FFNN) [55]: FFNNs process information in one direction, making them simple for time series forecasting. They use past values as inputs to predict future ones but might struggle with complex dependencies and long-term patterns.

DeepAR [56]: It is a deep learning model developed for probabilistic forecasting. It conditions predictions on the entire input time series, enabling it to capture complex patterns and provide uncertainty estimates, making it suitable for related time series forecasting.

N-BEATS ("Neural Basis Expansion Analysis for Time Series Forecasting") [57]: N-BEATS decomposes time series into interpretable basis functions, which it combines for forecasting. Its stacks and blocks design allows hierarchical feature extraction, providing flexibility, scalability, and interpretability for various forecasting tasks.

WaveNet [58]: WaveNet, originally for audio generation, is effective for autoregressive modeling in TSF. It models long-term dependencies, making it suitable for forecasting tasks where such dependencies are crucial.

Transformer [59]: The Transformer designed for Time Series is a model, with its self-attention mechanism, efficiently weighs the importance of different patterns of the input time series. It excels in capturing complex temporal patterns and dependencies, showing success in various sequence-to-sequence tasks, including time series forecasting.

5.3 Evaluation metrics

When dealing with time series datasets from various domains, choosing appropriate evaluation measures is critical to make sure that the forecasting models is assessed correctly. In the experiments of the papers MAE and RMSE are both scale-dependent metrics as the comparative analysis is between model time series of each dataset which means comparison is happening between homogeneous scales [60]. In this paper, MAE(Mean Absolute Error) is represented in Eq. 10 and RMSE(Root Means Squared Error) represented in Eq. 11 is used for the evaluation of the forecast.

$$MAE = \frac{1}{h} \sum_{i=1}^h |y_i - y_i^{pred}| \quad (10)$$

$$RMSE = \sqrt{\frac{1}{h} \sum_{i=1}^h (y_i - y_i^{pred})^2} \quad (11)$$

In Eqs. 10 and 11, h is the total number of forecasted points. MAE is calculated by the mean of the absolute differences taken between the actual values y_i and the predicted values y_i^{pred} in a time series. MAE is simple to compute and understand and also it is very less sensitive to outliers. RMSE calculates the square root of the mean squared deviations between the predicted values and the actual value y_i^{pred} and actual values y_i .

5.4 Hyperparameter tuning

The framework comprises three models; for each model, specific hyper-parameters are applied using grid search. The experiments necessitate the careful adjustment of several hyperparameters related to model training. The LSTM model in the TXtreme framework is composed of 4 layers followed by fully connected layers. Each LSTM layer is of size 100 LSTM cells and the last dense layer size equals the forecasting horizon for each dataset. The batch size for LSTM is 32 for all datasets and the learning rate is set to 0.001. As it is mentioned in section 4.3 loss function is set to Huber loss to counter the effect of any extreme values. The transformer model is trained on the oversampled data with extreme values. The total of encoder and decoder layers in the Transformer model is 2 which is the most common as more layers can capture more complex patterns but increase computational cost. Attention heads are set to 2 which is specifically used to focus on different parts of patterns as this paper hypothesizes two distributions (distribution of normal values and distribution of extreme values) head are set to capture both of them. As the forecasting horizon is 72 for all models, the output size is 72. The learning rate for the Adam [42] optimized is scheduled to change 1 % with an initial value of 0.001. Cross attention is applied to capture extreme values in time series with far distance.

Hyperparameters are carefully chosen to balance model complexity and computational efficiency in the FFNN(Feedforward Neural Network). In a (FNNN) designed to classify extreme values in time series, four hidden layers with 30 neurons in the first 3 layers each use ReLU activation functions, and the output layer is set to the size of the

forecasting horizon with sigmoid activation. A learning rate of 0.001 with a 1% change schedule, a batch size of 32, and several epochs with early stopping depending on the convergence are used. The Adam optimizer with He initialization [61] ensures efficient training.

For a Gaussian Mixture Model (GMM) to indicate extreme values score in time series data number of components ($n_components$) is set to 3, allowing the model to capture both data distributions. Initialization is done using k-means ($init_params$) for better starting points. The maximum number of iterations (max_iter) is set between 100 and 1000 to ensure convergence, with a convergence threshold around $1e-3$ to $1e-6$ for precise fitting. Regularization of the covariance matrix (reg_covar) with values from $1e-6$ to $1e-4$ ensures numerical stability.

6 Results and discussion

The predictive capacity and consistency of the proposed framework is assessed about state-of-the-art DNNs used for time series forecasting and are examined through three specific research questions outlined below:

- Research Question 1: Does this strategy generalize to datasets of multiple domains?
- Research Question 2: How effectively does our model adapt to and perform on previously unencountered data?
- Research Question 3: How accurate are the predictions for extreme values?

6.1 Research Question 1: Does this strategy generalize to datasets of multiple domains?

Table 4 documents the performance of the TXtreme framework which generalizes effectively across multiple domains, achieving superior or highly competitive performance in diverse datasets including solar energy, traffic, electricity and web traffic. TXtreme achieves the lowest RMSE and MAE, outperforming advanced models DeepAR, N-BEATS, FFNN, and WaveNet. In particular, it excels in Solar, Traffic, Electricity, and Web Traffic datasets. The underperformance of models like FFNN, DeepAR, N-BEATS, and WaveNet is due to their focus on general accuracy, which results in a smoothing effect that overlooks extreme values in time series data. For instance, in datasets with volatile or sporadic spikes, like Web Traffic Weekly, these models may fail to capture the full extent of extreme values, resulting in higher error values compared to the TXtreme. Additionally, while models like DeepAR and WaveNet incorporate advanced architectures for sequence prediction, their lack of specialized mechanisms for extreme value treatment means they may not handle extreme values as effectively as the TXtreme, which explicitly improves predictive accuracy and robustness across diverse and complex time series datasets.

TXtreme excels in datasets characterized by high variability and extreme values, providing an improvement ranging from 7.77% to 25.97%, but it may not have a noticeable impact in datasets with more regular patterns, such as the Traffic Dataset. TXtreme achieves a lower RMSE (6.10) compared to the closest competitor, N-BEATS (6.62), resulting in an improvement of approximately 7.85%. In terms of MAE, TXtreme (3.01) outperforms all other methods, with an improvement of about 8.23% over N-BEATS (3.28). The RMSE (484.89) for TXtreme is approximately 1.77% lower than DeepAR (477.99), and the MAE (244.23) for TXtreme shows a significant improvement of 25.97% compared to WaveNet (286.56). TXtreme outperforms other models with an RMSE (2614.13) that is 7.42% lower compared to N-BEATS (2820.62) and an MAE (1988.13) that is 11.69% lower compared to DeepAR (2272.58). However, for the Traffic Dataset, TXtreme shows comparable results to other methods (RMSE and MAE of 0.02). The limited improvement is because of low variability in

Table 4 Comparison of various models using RMSE and MAE for test data

Dataset	Evaluation matrix	FFNN	DeepAR	N-BEATS	WaveNet	Trans- former	TXtreme
Solar	RMSE	7.21	7.22	6.62	7.31	7.23	6.10
	MAE	3.28	3.28	3.52	3.55	3.28	3.01
Traffic	RMSE	0.02	0.02	0.02	0.03	0.03	0.02
	MAE	0.01	0.01	0.02	0.01	0.02	0.01
Electricity	RMSE	519.06	477.99	510.91	489.91	514.68	451.22
	MAE	354.39	329.75	350.37	286.56	398.80	244.23
Web Traffic	RMSE	2719.65	2981.91	2820.62	2719.37	3815.38	2614.13
	MAE	2025.23	2272.58	2051.30	2025.50	3100.32	1988.13

Table 5 Comparison of generalization ability of various models on unseen data on TXtreme

Dataset	Train RMSE	Test RMSE	RMSE % Difference	Train MAE	Test MAE	MAE % Difference
Solar	5.91	6.10	3.21	2.91	3.10	6.53
Traffic	0.02	0.02	0	0.01	0.02	0
Electricity	423.11	451.22	6.64	367.12	376.28	2.5
Web Traffic	2532.22	2614.13	3.23	2613.21	2633.13	0.76

The bold values in the "RMSE % Difference" columns indicate the percentage difference between train RMSE and Test RMSE

The bold values in the "MAE % Difference" columns indicate the percentage difference between train MAE and Test MAE

Bold values indicate low differences and the model generalizes to unseen data

Table 6 Results of individual components of the framework (TXtreme)

Dataset	Extreme RMSE	Normal RMSE	Total RMSE	Extreme MAE	Normal MAE	Total MAE
Solar	8.21	7.88	6.10	3.88	3.42	3.10
Traffic	0.03	0.03	0.02	0.03	0.03	0.02
Electricity	520.21	551.44	451.22	332.14	323.22	376.28
Web Traffic	2821.15	2781.11	2614.13	2722.56	2679.47	2633.13

Bold indicates RMSE and MAE of the hybrid model which is better in all cases

the data. The traffic dataset has a low RMSE and MAE across all models, which is due to fewer extreme values and that the data follows a more consistent pattern, which minimizes the advantage of TXtreme. If the dataset does not contain significant extreme events, the added complexity of TXtreme may not provide a meaningful advantage over simpler models like FFNN or DeepAR that already perform well with stable, non-extreme data.

6.2 Research Question 2: How effectively does our model adapt to and perform on previously unencountered data?

Table 5 provides a comparison of the TXtreme's generalization ability on unseen data across various datasets by examining both RMSE and MAE for training and test data, along with the percentage change between them. The consistent performance of the TXtreme can be attributed to an effective data sampling policy employed during training, which ensures the model is learning from varying patterns and extreme values, enhancing its robustness and ability to generalize well across various domains. For the Solar dataset, the TXtreme demonstrates excellent generalization with a low percentage change of 3.21% in RMSE and 6.53% in MAE, indicating only a slight increase in error from training to test data. The Traffic dataset maintains a consistent error with zero change, underscoring the model's robustness in highly stable environments. In the Electricity dataset, the RMSE change is 6.64%, and the MAE change is 2.5%, reflecting strong generalization despite the inherently volatile nature of electricity usage patterns. For Web Traffic, the changes are minimal, with 3.23% for RMSE and 0.76% for MAE, suggesting the model's effective handling of web traffic fluctuations.

6.3 Research Question 3: How accurate are the predictions for extreme values?

The Table 6 provides a detailed comparison of the TXtreme's performance on extreme and normal values across various datasets, highlighting its effectiveness in predicting extreme points. This enhanced performance is due to the integration of a classifier and an extreme value indicator within the model architecture. These components enable the model to identify and specifically focus on extreme events, thereby improving its predictive accuracy for such occurrences. For the Solar dataset, the TXtreme achieves an extreme RMSE of 8.21 and an extreme MAE of 3.88, contributing to a total RMSE of 6.10 and a total MAE of 3.10. This indicates the model's strong ability to handle both normal and extreme values effectively. For the Traffic dataset, the model maintains an identical RMSE and MAE of 0.03 for both extreme and normal values, indicating consistent performance across all data points. In the Electricity dataset, the Hybrid Transformer records an extreme RMSE of 520.21 and an extreme MAE of 332.14, leading to a Total RMSE of 451.22 and a Total MAE of 376.28.

This shows the model's proficiency in managing volatile electricity consumption patterns. In the Web Traffic dataset, the extreme RMSE is 2821.15, and the extreme MAE is 2722.56, which integrates into a total RMSE of 2614.13 and total MAE of 2633.13. The model's ability to predict extreme values is evident here, as it captures the significant fluctuations inherent in web traffic data.

7 Conclusion

This paper presents a novel framework called TXtreme, composed of multiple models, that better estimates rare but significant extreme values in time series to reduce overall forecasting errors. TXtreme captures extreme values that generally range between 5% to 15% in datasets, improving forecasting accuracy by 7% to 25%. Overall TXtreme model demonstrates exceptional generalization across multiple domains, effectively predicting extreme values and unseen data. It achieves superior or highly competitive performance in diverse datasets such as solar energy, traffic, electricity and web traffic, often outperforming other advanced models by addressing extreme values in time series data. The model's robust performance on unseen data, evidenced by minimal percentage changes in RMSE and MAE between training and test sets, highlights its ability to handle various patterns and anomalies. Additionally, its integration of a classifier and extreme value indicator significantly enhances its accuracy in predicting rare and significant events.

In future, it is interesting to investigate TXtreme for multivariate time series forecasting data, enhancing forecasting accuracy by capturing interrelated dependencies among multiple variables. Additionally, developing a unified transformer architecture that integrates LSTM, feed-forward neural networks along with specialized attention mechanisms tailored for extreme value forecasting would further enhance the model's sensitivity to critical patterns associated with extreme values.

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